

Subject-Aware Multimodal Human Action Recognition under a 100 MB Budget: A 96 MB Strideless R(2+1)D Ensemble with a Recording-Structure Decoder

Technical Report — CUHK-X Challenge 2026, Small Model Track (UbiComp/ISWC 2026)

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Abstract

We present a 96 MB four-member ensemble of quantized R(2+1)D-34 video models [2] for the CUHK-X Challenge 2026 Small Model Track. It scores **0.89552** (180 of 201 scored clips) on the public leaderboard, 11th of 326 teams, and **0.90196** (184 of 204 clips) on the private leaderboard, 8th. The honest cross-subject number — the one that predicts the organizers’ 30%-weighted evaluation on fresh participants — is our nested five-fold, subject-held-out estimate over all 3,036 training clips: **0.7770** raw per-clip and **0.8521** after the structure decoder.

Two structural findings about the challenge data drive the design. (i) The public test set is *not* fully cross-subject: 275 of 405 test clips (68%) were recorded on a date when the same user also appears in training, so leaderboard feedback systematically over-reports cross-subject skill, and we train the deployment model on all 18 subjects while validating on held-out subjects. (ii) Recordings are scripted, with actions repeated in takes whose boundaries are recoverable from sensor metadata alone. We exploit the second property with a label-free recording-structure decoder worth **+7.5** pooled points on held-out subjects (+4.0 to +9.8 per fold), guarded per take so that the worst case is bounded and data without the structure degrades exactly to raw per-clip argmax — verified on a metadata-stripped copy of the test set, and measured on the leaderboard itself (0.89054 $\rightarrow$ 0.85074 when the decoder is switched off). The ensemble combines a joint body+hands member with a cross-view interaction transformer, a depth+IR member, a thermal member and an IR member, with per-clip missing-modality masking. All shipped model assets total 96,081,702 bytes under the 100 MB cap; inference is deterministic, needs no network access and no training data, takes 83 s for 405 clips at 1.71 GB peak GPU memory, and reproduces the submitted CSV byte-for-byte.

1 Introduction

The CUHK-X Challenge 2026 [1] evaluates human activity recognition (HAR) from privacy-preserving, non-RGB sensing modalities. In the Small Model Track, a short clip with six synchronized modalities — infrared (IR) video, RGB-D depth, thermal video, IMU from five wearable sensors, a skeleton stream, and mmWave radar — must be classified into one of 40 activity classes. Scoring is strict top-1 accuracy on the primary intended action. The track imposes a hard budget: model size at most 100 MB, conventional architectures (CNN/RNN/Transformer) only, no large pretrained

backbones, and no closed-source APIs or LLMs in development or labeling. The final ranking blends the Kaggle private leaderboard (20%), an organizer evaluation on eight held-out participants (30%), reproducibility (10%), the technical report (20%), presentation (10%), and model efficiency (10%).

Our submission, SOTA-ENSEMBLE, is 11th of 326 teams on the public leaderboard and 8th on the private leaderboard. Because 30% of the final score comes from fresh participants, we lead with the cross-subject numbers rather than the leaderboard ones throughout (§3 explains why), and we are explicit about which components we expect to transfer and which we do not. Four design decisions dominate the result:

1. **A data-aware training split.** The test-set construction turned out to be partly same-subject, same-day (§2). We therefore train the shipped models on all 18 subjects, but keep held-out-*subject* validation as the honest generalization metric, because the organizer’s 30%-weighted evaluation is cross-subject.
2. **A strideless temporal recipe.** The R(2+1)D-34 backbone compresses its input to two temporal positions before classification. Setting six temporal strides to 1 — not a learned parameter, and zero extra bytes — keeps every position and, together with two other recipe changes we report honestly in §4.2, improves held-out-subject accuracy by +3.6 decoded points.
3. **A label-free structure decoder with a guard.** Because recordings are scripted repeats, take-level consistency is a free signal. A decoder that groups clips into takes from sensor metadata and refines the ensemble probabilities under a distinctness constraint adds +7.5 pooled points on held-out subjects; a per-take guard bounds the worst case when the structure is absent or incoherent (§4.4).
4. **A validation protocol built around a measured leak.** Most of our development time went into knowing which numbers to trust: a separation profile that exposes a leaky split, a measured noise floor, and a decoder null test that shows several of our own rejections were under-powered (§3).

The report is organized as follows. §2 describes the data and the structural findings that drive the design. §3 fixes the validation protocol. §4 describes the unified model: members, interaction transformer, quantization, and the decoder (Figure 2 shows the full inference flow). §5 gives the complete training recipe. §6 reports results, ablations and an error analysis. §7 covers reproducibility and efficiency; §8 data usage and rule compliance; §9 limitations and future work.

2 Task, Data, and Structural Findings

2.1 Task definition

Given one clip per test sample, we predict the primary activity label from 40 classes. The 405-row public test set is split by Kaggle into 201 clips scored on the public leaderboard and 204 scored on the private leaderboard; every public score is an exact multiple of $1/201$ and every private score an exact multiple of $1/204$, which is how we recovered the split sizes. Each clip ships six synchronized modalities; the two video modalities we find most useful are IR (blank in 4 test clips) and thermal (blank in 10), while depth is present in all.

2.2 The test set is not fully cross-subject

Clip filenames carry absolute timestamps
(Depth_2025-06-01_16-01-45.916_00000180_Color.png) and the IMU CSVs carry device

grouping	fold-0 acc.	verdict
held-out subject	0.6111	pessimistic about the LB: models only the unseen-session third
held-out block	0.8386	leaky : leaves a retake ≈ 70 s later in training
held-out scene region (gap 600 s)	0.7027	matches the real test construction

Table 1: Cross-validation groupings, same recipe and fold index, on fold 0. These are *early single-model, 16-frame* numbers used to choose the protocol; they are not the shipped ensemble, whose cross-subject accuracy is in Table 9. The held-out-block protocol is optimistic because real test blocks are recorded days after their training neighbours.

MAC addresses. Extracting these for all 3,441 clips shows the test set was not drawn by holding out subjects:

- Training covers 12 recording dates; the 405 test clips fall on 7 dates.
- Five of those dates are shared with training (2025-05-31, 06-01, 06-02, 06-12, 06-13). **275 of 405 test clips (67.9%)** were recorded on a date when that same user was being recorded for the training set.
- No test clip’s time window overlaps a training clip’s, so the split is by whole recording block — not by subject, and not by time.
- Only 130 test clips (2025-06-16 and 06-17) come from sessions with no training data at all.

Two consequences follow, and both shape the rest of this report. First, grouping cross-validation on **user** measures only the hardest $\approx 32\%$ of the test set and applies that estimate to all of it, so it is systematically *pessimistic about the leaderboard*. Second — and more important for the final ranking — *leaderboard score is a biased estimate of cross-subject skill*, in the optimistic direction. We therefore report both, and treat the held-out-subject estimate as primary.

2.3 Recordings are scripted

A *block* is one continuous take of a scripted action sequence; a *scene* is that block repeated 2–3 times. In the labelled data every action inside a block is distinct (780/780 blocks), and 79.5% of scenes repeat the exact action order (209/263; 82.9% by set). Blocks are recoverable from metadata alone: the thermal frame counter (`frame_000861.jpg`) resets at each take and runs at 24 fps, so a block is a maximal run whose counter tracks the wall clock. Grouped additionally by date and IMU device set, **890/890** recovered blocks are *pure* against ground-truth (date, user, trial), and all 890 satisfy distinctness. On the test set the same procedure recovers 165 blocks; 355 of 405 clips sit in a block of size ≥ 2 and 269 sit in a scene with at least one sibling repetition. This is what the structure decoder in §4.4 exploits. It uses recording metadata, not the activity signal; it is inherent in the released test data and needs no external information, but it is a separate decision from the split fix and is kept switchable in the code.

3 Validation Protocol

We tried three cross-validation groupings; only one mirrors the real test construction (Table 1, Figure 1).

The **separation profile** is the diagnostic that settles it: the gap in session time from each held-out block to the nearest block still in training.

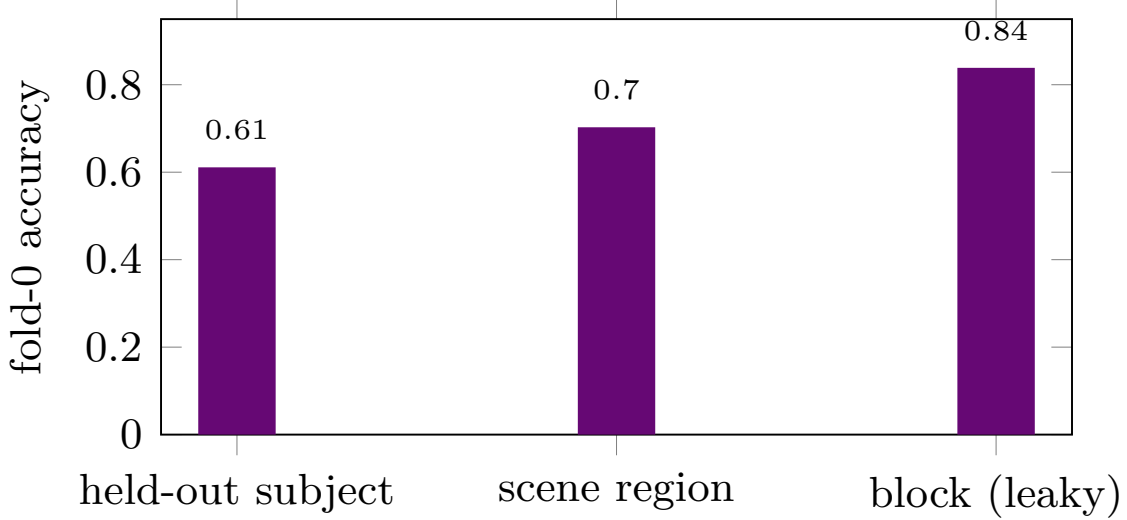


Figure 1: CV grouping accuracies on fold 0 (from Table 1). The block-held-out protocol is the *leaky* one: its median separation to training blocks is 70 s versus 3,793 s for the real test set, so it inflates accuracy by leaving near-duplicate retakes in training.

held-out set	min	p10	median
real test set	286 s	831 s	3,793 s
block-held-out CV	≈ 0 s	—	70 s
scene-region CV (gap 600 s)	609 s	979 s	2,732 s

Table 2: Separation profiles. We check every new split’s profile against the test set before trusting a number computed on it.

Held-out-*subject* folds remain our primary generalization metric, because the organizer’s 8-participant evaluation is cross-subject. The development split is fixed and hashed (869ec27d...0e835): 18 subjects, four development folds holding out 3–4 subjects each (530/540/696/640 rows over 14 subjects, 2,406 rows total), plus a fifth fold scoring the four formerly locked confirmation subjects (user3, user5, user17, user23; 630 rows) with models trained on all development rows. Every training set excludes its own validation subjects, and every decoder prior is fit on the corresponding training fold only.

Three further rules keep the protocol honest.

The noise floor. Fold 0 is 530–619 rows depending on the cache generation; differences below about **3 points are not reliable** on a single fold. Evidence: one member moved 2.6 points ($0.7577 \rightarrow 0.7318$) purely from changing a quantization group size from 128 to 512, which should be near-neutral. A 160px submission made on a +0.8 CV difference lost 3.0 points on the leaderboard. We do not spend a submission on a sub-3-point single-fold CV difference.

Four check levels and an interval. Every architectural change must be positive at four check levels (development raw/decoded and locked-subject solo/in-ensemble) and carry a 95% subject-bootstrap confidence interval [13] — resampling *subjects*, not clips, since clips within a subject are strongly dependent — before it ships. We state the number of subjects resampled with every interval, because a cluster bootstrap over few clusters is fragile: the strideless interval (§4.2) resamples only **6** subjects, the four-fold intervals resample **14**, and the five-fold intervals resample all **18**.

	fold 0	fold 1	fold 2	fold 3	fold 4
null decoded Δ (mean $\pm$ sd, clips)	-6.4 ± 5.9	-7.1 ± 3.6	-2.5 ± 4.4	-1.9 ± 4.0	$+0.6 \pm 6.4$
times the fold was used to decide	most	most	confirmation	confirmation	once

Table 3: Decoder null test: decoded accuracy change from noise matched to a real swap’s argmax flip rate. Fragility tracks decision reuse. Judging a candidate against zero, rather than against this null, biases against every candidate on folds 0/1.

A decoder null test, which corrected us. Late in development we measured what the decoder does to a *null* perturbation: Gaussian noise on the base combined log-probabilities, scaled so the raw argmax flip rate matches that of a real candidate swap, 30 draws (Table 3). On folds 0/1 the expected null decoded delta is ≈ -13.5 clips. Several candidates we rejected for “breaking the decoder” lost -7 to -17 clips on those folds — i.e. inside or barely outside the null. Folds 0/1 were used to choose the decoder, the fusion weights and every accepted member, so their decoded score is overfit there (a winner’s curse) and *any* perturbation regresses. We record this because it means those rejections are weak evidence, not strong evidence, and because fragility tracked exactly how often a fold had been used for decisions.

4 Method

4.1 The unified model: full inference flow

Figure 2 shows the complete inference pipeline. A raw clip is detected (YOLO11n person box + YOLO11n-pose wrists [6], per clip, no cached boxes), cropped and cached into per-view inputs, scored by four parallel R(2+1)D-34 members with missing-modality masking, fused by a fixed weighted average, and finally refined by the label-free structure decoder with its per-take guard. Everything runs offline from the raw clip folders.

4.2 Backbone and the strideless temporal recipe

All four members are R(2+1)D-34 video encoders [2] with 63,514,700 parameters, fine-tuned from the IG65M→Kinetics clip-32 initialization [3, 4]. With 16 input frames, the stock backbone produces $T=8$ positions after layer 2, $T=4$ after layer 3, and $T=2$ after layer 4: six temporal convolutions with stride-2 downsampling in layers 2–4. Because stride is a property of the *convolution*, not a learned parameter, setting those six strides to 1 uses the identical pretrained filters, keeps every temporal position, and changes neither the parameter count nor the checkpoint size. We call this configuration *strideless*. It is free in bytes but not in compute: at 32 frames and 128 px it costs 446.75 GMACs per encoder pass against 200.33 GMACs for the strided backbone at the same length, a factor of 2.23 (§7).

Validated against the strided IR baseline on held-out subjects, all four check levels are positive (Table 4); the pooled decoded gain is **+0.0355** with 95% subject-bootstrap CI [+0.0056, +0.0647]. The structure decoder also gains *more* with strideless inputs (+7.9/+6.3 decoded points versus +5.9/+4.8 for the strided baseline): finer temporal evidence yields better-calibrated probabilities.

Attribution caveat, stated because it matters. A later replay of the two trainers showed they are *not* a pure stride ablation. The strideless trainer also omits the baseline’s mixup [10] ($\alpha=0.2$) and its two-epoch encoder-freeze warmup. The recipe reproduces exactly and the gain is real, but the +0.0355 is the effect of the *recipe* — stride removal, no mixup, no freeze warmup — and cannot be attributed to stride removal alone. We discovered this mid-experiment and chose to

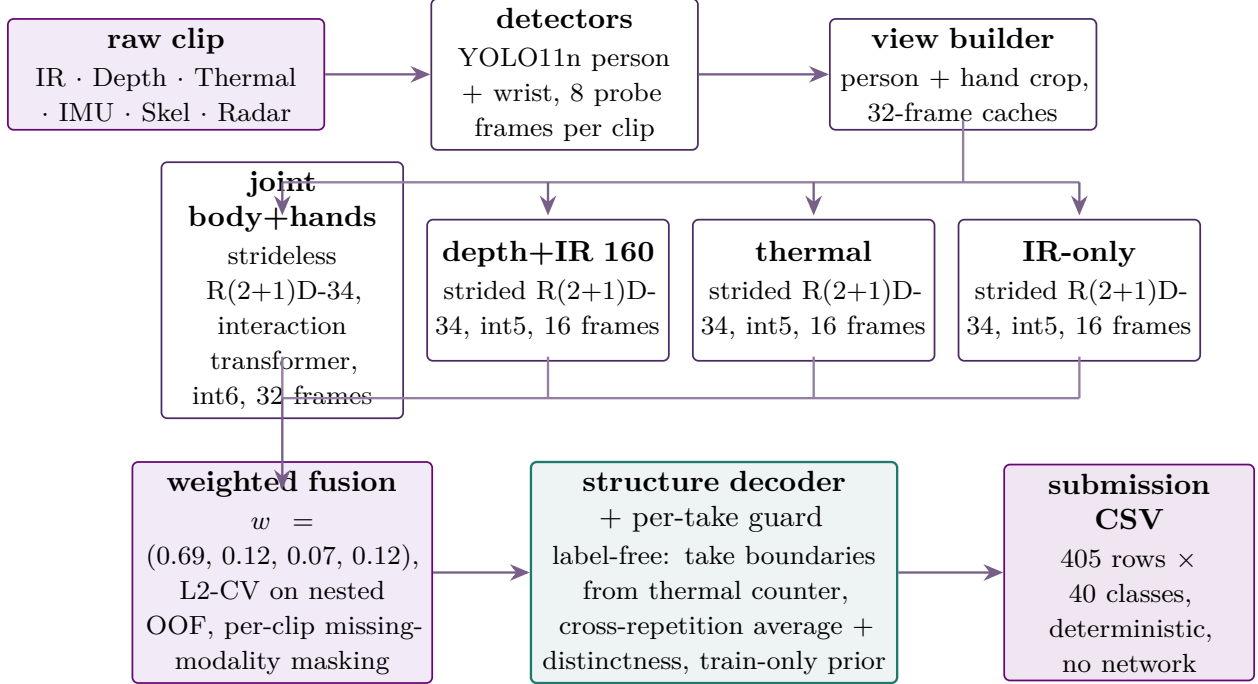


Figure 2: The unified model, end to end. Stage 1 (top): per-clip detection and view construction. Stage 2 (middle): four parallel R(2+1)D-34 members; the joint body+hands member is the only strideless one and carries a 2-layer cross-view interaction transformer. Stage 3 (bottom): fixed weighted fusion with per-clip missing-modality masking, then the label-free structure decoder with its per-take guard, which degrades to raw argmax when the take structure is absent or incoherent. All steps run offline from the raw clip folders; no training data, no cached boxes, no network access.

preserve the recipe for the remaining comparisons rather than silently change the baseline. A clean one-factor ablation is the first thing we would run with more time.

Configuration note. Table 4 is a **16-frame** comparison (`gen_stride16_mean_f{0,1}_s3407`). The shipped strideless member runs at **32** frames; the additional gain from 32 frames is measured separately in Table 10.

Two non-architectural hazards are handled explicitly. A strideless checkpoint is *shape-identical* to a strided one and loads without error into the wrong architecture; the stride flag therefore travels with the package and is applied at inference, with the byte-for-byte raw-reproduction check as the safety net. And the temporal *mixer* is not the cause: a temporal convolutional head beat mean pooling on frozen features but lost end-to-end, and a Mamba-style SISO state-space stack lost to plain mean pooling on both folds at 50× the parameters, so plain mean pooling ships.

4.3 The four members and fusion

Joint body+hands. A single shared strideless R(2+1)D-34 encodes both the person crop and a tight hands-only crop derived from YOLO11n-pose wrists (median zoom 2.31×; wrists found in 95.7% of clips, person-crop fallback otherwise). A 2-layer interaction transformer (width 256, 4 heads, 1,206,568 parameters, pre-norm) fuses per-frame body and hands tokens, with zero-initialized logits added to the fused head and per-view auxiliary heads at weight 0.25. This is the diversity win: hands/objects are where the remaining confusions live (§6.3).

Missing-modality masking. Thermal is blank in 10/405 test clips and IR in 4; blank members

check	fold 0	fold 1
development, raw TTA	+0.0283	+0.0074
development, decoded	+0.0491	+0.0222
locked subjects, solo	+0.0286	+0.0318
locked subjects, in ensemble	+0.0079	+0.0349

Table 4: Strideless recipe vs. the strided IR baseline, **16 input frames**, held-out subjects. Pooled decoded gain +0.0355 (1,070 rows), 95% subject-bootstrap CI [+0.0056, +0.0647] resampling the **6** held-out subjects of folds 0–1. See the attribution caveat in the text.

member	view	cache	frames	strideless	bits	fusion w
joint body+hands (w69)	IR	ir128 + irhand128	32	yes	int6	0.69
depth+IR 160	combined	union32 ₁₆₀	16	no	int5	0.12
thermal	combined	thermal32	16	no	int5	0.07
IR-only (strided)	IR	union32	16	no	int5	0.12

Table 5: Ensemble members of the primary submission. Weights are fit on nested out-of-fold predictions with L2 chosen by inner leave-one-fold-out on raw NLL, then frozen.

are dropped per clip and the remaining weights renormalized, so no member ever scores a clip it cannot see.

Selection by evidence diversity. The ensemble stays IR-led with thermal as the complementary sensor — thermal wins bathroom actions, IR wins floor/table/fitness actions. Adding a second 32-frame IR member raised raw accuracy by 1.0–1.6 points but lost 7–17 decoded clips on folds 0/1, and we rejected it on that basis. As Table 3 shows, that loss is inside the null band for those folds, so the rejection is *weak* evidence; the defensible statement is that the candidate showed no reliable decoded gain and the incumbent was kept.

4.4 Recording-structure decoder

The decoder consumes only metadata. For each clip it recovers take (block) boundaries from the thermal frame counter and timestamps, groups clips into takes and scenes, aligns partial repeats within a take, applies a cross-repetition average (strength 0.3) plus a distinctness assignment — one action per take, solved as a linear assignment [7] over a beam of 512 candidates — under a train-only action-transition prior, and outputs one label per clip. On held-out subjects it is worth **+7.5 pooled points** (Table 9), and more on subject folds than on scene folds, which is the right direction for a cross-subject organizer evaluation.

Per-take guard. The decoder’s failure mode is a take that is too large or incoherent: the distinct-action constraint then forces wrong labels. A stress test quantifies it (Table 6). The guard — fall back to raw per-clip argmax when a take exceeds 10 clips (the training maximum) or contains more than 2 confident ($p \geq 0.5$) same-class pairs — recovers most of the collapse at a cost of 7 clips in 3,036 on clean data. On the real 405-clip test set, 1 of 151 recovered takes fell back.

Honest degradation. This component is the one most likely to degrade on unseen participants, because the public set’s 68% same-user, same-day clips may not recur in fresh data. We bound the exposure three ways, and we have measured all three rather than argued them:

1. **The raw base is strong on its own:** 0.7770 pooled cross-subject, so the decoder is a bonus, not the load-bearing wall.

condition	decoded	guarded	vs. no decoder
clean (as shipped)	85.54%	85.31%	—
no thermal counters (takes rebuilt from timestamps)	83.17%	83.17%	+120 clips
takes split	82.61%	82.61%	+103
order shuffled within take	79.15%	—	−2
singletons with timestamps	78.43%	—	−24
random groups	73.75%	74.80%	−166
takes merged	58.63%	72.13%	−625 → −215
no structure at all (= raw argmax)	79.22%	—	—

Table 6: Decoder stress test on nested out-of-fold probabilities, 3,036 clips. The guard costs 7 clips on clean data and recovers 410 of the 625 lost to merged takes.

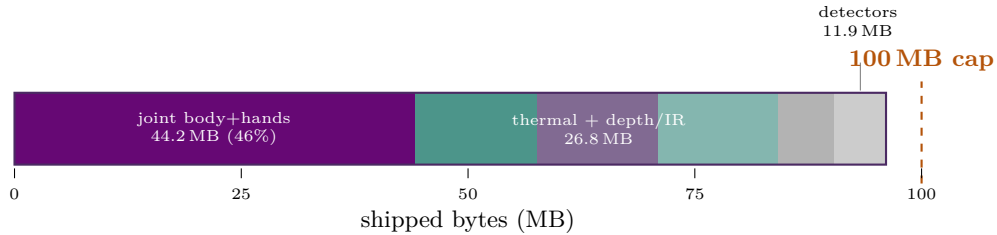


Figure 3: The 100 MB budget, spent. Stacked bar of the six shipped *weight* files (Table 7): joint member 44.2 MB, thermal 13.5, depth+IR 160 13.4, IR-only 13.2, pose detector 6.3, person detector 5.6. The seventh asset, the 0.07 MB transition prior, is included in the total but too small to plot. Total 96.08 MB, 96.1% of cap.

2. **The guard bounds the worst case** (Table 6).
3. **With no structure the pipeline degrades exactly to raw argmax.** We built `robust_test_stripped`: hardlinked copies of the test clips with IR/Depth timestamps and thermal counters *renamed away*, pixels identical, so `clip_meta` finds no t_0 or thermal origin in any clip. Fresh inference completes normally, every clip becomes a singleton block, combined probabilities are identical to the structured run (max absolute difference 0) and labels equal raw argmax for 405/405 rows. This is the metadata-stripped fallback test, not merely a shuffle.
4. **We measured the fallback on the leaderboard itself.** A diagnostic submission of the no-structure output of the secondary model scored 0.85074 (171/201) against 0.89054 (179/201) structured: the decoder is worth 8 public clips, −4.0 points, a 4.5% relative drop. It is *smaller* on Kaggle than in CV precisely because 68% of the test clips are seen sessions.

4.5 Quantization and the 100 MB budget

All four members are group-quantized (group 128; int6 for the joint member, int5 for the rest) and losslessly lzma-delta-coded against the first member; every decoded tensor round-trips exactly against an independent quantization of the same model. 3-bit quantization collapses to chance at every group size tried, and 4-bit costs more accuracy than the byte budget buys back, so 5/6 bits is the operating point. The shipped assets and their budget share are in Table 7 and Figure 3. Inference runs in fp16 with horizontal-flip TTA [12]; the IG65M checkpoint is *not* loaded at inference (models are built `pretrained=False`), so there is no download and no large backbone in the shipped package.

asset	role	quant.	bytes
video_gen_joint_interact_all_s3407.pt	joint body+hands member	int6, group 128	44,171,513
video_all_r_thermal.pt	thermal member	int5, group 128	13,456,643
video_all_r160.pt	depth+IR 160 member	int5, group 128	13,354,151
video_all_r_ironly.pt	IR-only member	int5, group 128	13,164,119
yolo11n-pose.pt	wrist detector	fp32	6,255,593
yolo11n.pt	person detector	fp32	5,613,764
<i>subtotal, six weight files</i>			96,015,783
sequence_prior.json	action-transition prior	—	65,919
total shipped model assets			96,081,702

Table 7: The seven shipped inference assets of the primary submission (measured bytes; `model_asset_bytes` in `report.json`). Members 2–4 are lzma-delta-coded against the first member. Total 96,081,702 B = 96.08 MB, 96.1% of the 100 MB cap.

5 Training Details

Every shipped member is trained with the same optimizer and schedule (Table 8); the members differ only in input cache, frame count, stride configuration and the two options noted in the table footnote. Training and inference both run on a single RTX 5090.

Preprocessing. Person boxes come from YOLO11n run on 8 frames spread evenly across each clip’s IR stream; the union of those boxes, expanded by a margin of 1.40 and squared off, defines the crop. Frames are decoded at 128 px (160 px for the depth+IR member), missing images stay black, and caches store 32 frames so training can draw a random temporal window. The hands view uses YOLO11n-pose wrists plus a half-forearm extension, padded 8%, minimum side 96 px, falling back to the person crop where no confident wrist is found.

Fusion weights. Members are combined by a fixed weighted average of probabilities. The weights are fit on nested out-of-fold predictions — fit on four folds’ OOF rows with L2 strength chosen by inner leave-one-fold-out on raw NLL, then evaluated on the held-out fold — so the reported ensemble numbers do not include weight-fitting optimism. The final shipped weights (0.69, 0.12, 0.07, 0.12) are the refit over all 3,036 OOF rows; per-fold fitted weights range 0.60–0.74 for the joint member, confirming the fit is stable.

6 Results

6.1 Headline

Table 9 is the number we would want a judge to read first. The pooled cross-subject estimate, 0.7770 raw and 0.8521 decoded over all 18 subjects, is what we expect to carry to the organizers’ fresh participants; the leaderboard scores are higher because 68% of that test pool is same-user, same-day (§2.2). The decoder is worth +7.5 pooled points, ranging from +4.0 (fold 4) to +9.8 (fold 0).

A second Selected submission, `release_joint_ir4_w64` (Kaggle ID 56229279), ships the same pipeline with the original linear joint head and refit weights (0.64/0.14/0.07/0.15) as a hedge; both Selected models reproduce byte-identically from the verification package. The private score (0.90196) slightly exceeds the public score (0.89552). We note this but do *not* claim it as evidence against

setting	value
initialization	IG65M→Kinetics-400 R(2+1)D-34 clip-32 [3]; 4-channel stem, RGB weights copied, 4th channel = their mean
optimizer	AdamW [8], weight decay 0.01
learning rates	encoder 1×10^{-5} ; head 1×10^{-3} ; interaction transformer 3×10^{-4}
schedule	2 warmup epochs (linear), then cosine decay to $0.05 \times$
epochs	40
batch	8, gradient accumulation 2 (effective 16)
loss	cross-entropy, label smoothing 0.05 [9]; joint member adds $0.25 \times (\text{body CE} + \text{hands CE})$
gradient clipping	global norm 5.0
precision	fp16 autocast + GradScaler (init scale 1024) [12]
BatchNorm	pretrained statistics frozen (eval mode) throughout
seeds	3407 (joint, IR-only), 44 (depth+IR 160), 33 (thermal)
<i>augmentation (training only, per-clip RNG seeded by (seed, epoch, row, view))</i>	
horizontal flip	$p = 0.5$
temporal roll	$p = 0.5$, shift $\in \{-1, 0, +1\}$ frames
zoom crop	$p = 0.5$, scale $\sim \mathcal{U}(0.75, 1.0)$, random position, bilinear resize back
brightness	multiplicative $\sim \mathcal{U}(0.85, 1.15)$, clamped to $[0, 1]$
random erasing [11]	$p = 0.25$, box 10–30% of each side, filled with 0
frame sampling	16-frame members: stride-2 with per-frame random offset jitter; 32-frame members: all cached frames in order
mixup [10]	$\alpha = 0.2$ for the three strided members; <i>off</i> for the strideless recipe
encoder freeze	first 2 epochs for the three strided members; <i>off</i> for the strideless recipe
<i>cost (single RTX 5090, 24 CPU cores, all-Training fits)</i>	
joint body+hands	178.3s/epoch, 118.8 min total, 26.6 GB peak GPU
IR-only	88.0s/epoch, 58.7 min total
depth+IR 160	43.4s/epoch, 29.0 min total
thermal	28.0s/epoch, 18.7 min total
four shipped members	3.8 GPU-hours (excluding cache construction and all rejected runs)

Table 8: Complete training recipe. Model selection during development uses best validation accuracy on the held-out subjects of the fold; the shipped all-Training fits have no validation set and take the final (40th) epoch.

test-set overfitting: both halves are drawn from the same structurally leaky pool, so they are not independent in the way that argument would require.

6.2 Ablations

Two clarifications the raw table cannot carry.

The split fix. The submission that introduced training on all 18 subjects also introduced the structure post-processing, and the public score moved 0.66666 (134/201) $\rightarrow$ 0.71144 (143/201): +9 clips, +4.5 points. We cannot cleanly separate the two contributions from that submission, and we do not try to. The number is on the public leaderboard, not on held-out subjects, and the decision it justified — *train on every subject, validate on held-out ones* — is the one that matters.

Everything else is small. Once the split and the decoder were in place, every remaining accepted change was worth 0.6–3.6 points, most of them with intervals that nearly touch zero. That is the honest shape of the result: two structural decisions, then a long tail of marginal gains, several of which we would not defend individually.

	rows	raw per-clip	decoded
fold 0 (user2, user24, user6)	530	0.7415	0.8396
fold 1 (user20, user22, user9)	540	0.8019	0.8889
fold 2 (user16, user18, user19, user4)	696	0.7830	0.8736
fold 3 (user1, user21, user7, user8)	640	0.7531	0.8172
fold 4 (user3, user5, user17, user23)	630	0.8032	0.8429
pooled, all 18 subjects	3,036	0.7770	0.8521
public leaderboard (201 scored clips)	201	0.85074 [†]	0.89552 (180/201)
private leaderboard (204 scored clips)	204	—	0.90196 (184/204)
chance (40 classes)	—	0.025	—

Table 9: Primary submission `release_joint_interact_w69_guard` (Kaggle ID 56239764). The five fold rows are the *nested* cross-subject estimate with fixed fusion weights: every model excludes its validation subjects and every decoder prior uses only the corresponding training fold. [†]Measured with a separate diagnostic submission of the secondary model’s no-structure output (171/201), against that model’s 179/201 structured. Public and private leaderboards are two disjoint halves of the *same* 405-clip test pool described in §2.2 — neither is a held-out-user set.

change	protocol	rows	net clips	points	95% CI (subj.)
split fix + structure	public LB	201	+9	+4.5	—
strideless recipe (16 frames)	2 folds	1,070	+38	+3.6	[+.0056, +.0647] (6)
32 frames for the IR member	4 folds	2,406	+30	+1.2	[+.0016, +.0228] (14)
joint body+hands member	5 folds	3,036	+38	+1.3	[+.0026, +.0231] (18)
balanced class sampling	4 folds	2,406	+23	+1.0	[+.0030, +.0160] (14)
interaction transformer (joint member)	5 folds	3,036	+18	+0.6	[−.0027, +.0149] (18)
thermal member	5-fold OOF, scene	3,036	—	+2.3	—

Table 10: Ablations that moved the score, decoded accuracy on held-out subjects unless noted. “Net clips” is fixed minus broken over the stated rows; “points” is that divided by the row count. Intervals are subject-bootstrap, with the number of resampled subjects in parentheses. The interaction transformer’s interval includes zero; it shipped because it also lowered NLL on 4 of 5 folds and combined with the weight refit for a larger joint effect.

6.3 Where the remaining error is

Ensemble top-1 is 0.7415 on fold 0, but top-2 is 0.8352 and top-3 0.8708: the correct answer is almost always inside a tiny candidate set. The errors concentrate in close-range hand/object actions (Table 11); perfectly resolving the eight largest confusable pairs would be worth **+7.4 points**.

The diagnostic that explains why we stopped attacking these is in Table 12: evaluated without augmentation, a fold checkpoint separates the confusable classes at 0.97–0.99 accuracy on clips it has *seen* and at 0.56–0.83 on clips from new scenes. This is a *generalization* gap, not a capacity gap. Fine-tuning toward the separation cannot help when the training signal is already saturated, which is why every specialist, resolution and self-training attempt below failed. Complementary *sensor* evidence — thermal, hands — rather than more spatial or temporal resolution of the same sensor, is the lever that still works.

7 Reproducibility and Efficiency

The submitted package runs `inference.sh <data_dir> <output_csv>` from its root (or an equivalent one-command `docker run` with `-network none`). On the 405-clip public test set it reproduces

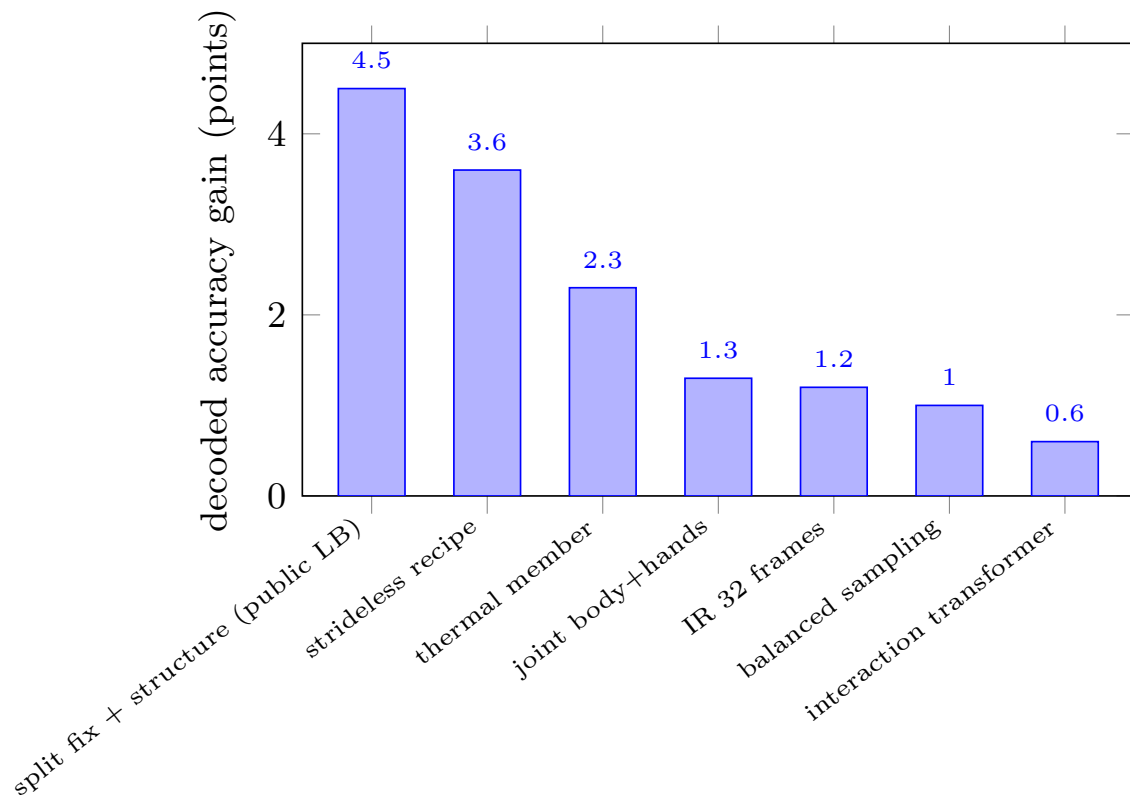


Figure 4: What moved the score, in accuracy *points* on the protocol stated for each change in Table 10. The first bar is a public-leaderboard measurement and bundles two changes; the rest are held-out-subject measurements. Gains are not additive and the axis is shown only to convey relative magnitude.

the submitted CSV **byte-for-byte** (SHA-256 f77d1d28...ebc3893) in **83.0 s** at **1,713,850,880 B** (1.71 GB) peak GPU memory on a single RTX 5090, with per-clip YOLO detection and no cached boxes. The pipeline is deterministic on a fixed GPU and driver: no random numbers are drawn at inference.

Efficiency is not only bytes. Table 14 reports parameters and multiply-accumulates alongside file size, because a 96 MB package that runs four 63.5 M-parameter networks is not cheap at inference, and one of those networks is strideless, which multiplies late-layer compute by $2.23\times$ at equal frame count. Competitors with genuinely small architectures will and should score better on this axis; we report the figures rather than let file size stand in for them.

8 Data Usage and Compliance

We declare everything below explicitly rather than rely on interpretation.

- **Small Model Track Training and Testing sets:** used as released. All 3,036 Training clips enter the shipped all-data fits.
- **No test-set labels** in any form: not for training, model selection, decoding, or validation. All decoder priors are fit on Training labels only, with fold-specific priors for every validation number in this report. We are aware of a public Hugging Face mirror of this competition’s data

true class	predicted class	out-of-fold errors
Turn_pages	Read_documents	11
Take_and_use_tableware	Stir_drinks	9
Play_games	Take_a_selfie	4
Wipe_hands	Wipe_bowls	3
Write	Tap_the_keyboard	2

Table 11: Largest residual confusion pairs, out-of-fold. Six confusion clusters hold 257 of 645 out-of-fold errors.

cluster	fold	held-out	training rows
desk (Read_documents, Turn_pages, Write, Tap_keyboard)	0	0.565	0.990
face (Eat, Drink, Brush_teeth, Take_medicine, ...)	0	0.827	0.983
desk	1	0.681	0.968
face	1	0.631	0.978
desk	2	0.604	0.967
face	2	0.793	0.980

Table 12: Within-cluster accuracy before any specialist training. The backbone already separates these classes almost perfectly on seen clips and poorly on new scenes: the distinguishing cues are specific to the training recordings.

whose `test.csv` may contain real test labels; we read only its header and adopted a standing rule never to read or use its values.

- **Leaderboard feedback:** the public leaderboard *was* used as a decision signal, in the ordinary way — to confirm the split fix and to choose between candidate releases. No test labels were used, but we state this plainly rather than claim that no test-set information influenced development.
- **Large Model Track data:** permitted by the organizers for pre-training/co-training in this track. We audited it for leakage (it contains only the 18 Small-Track users; test users appear nowhere) and did *not* use it in any shipped model — it added no new subjects and almost no new labeled action footage. The package is reproducible without it.
- **Initialization.** The R(2+1)D-34 members are initialized from IG65M→Kinetics-400 pretrained weights [3, 4] and then fine-tuned end-to-end on challenge training data. The weights reach us through the MIT-licensed conversion in `moabitcoin/ig65m-pytorch` [5], whose license ships in the package. R(2+1)D-34 is a conventional 3D CNN of 63.5 M parameters; it fits the 100 MB cap after quantization, and the pretrained checkpoint is neither shipped nor loaded at inference (models are built `pretrained=False`), so the package carries no pretrained backbone and needs no download.
- **Detectors — corrected declaration.** The person and wrist detectors are the *stock, off-the-shelf* Ultralytics YOLO11n and YOLO11n-pose release weights [6], COCO-pretrained (checkpoint metadata: `coco.yaml` / `coco-pose.yaml`, released 2024-09-25). They are *not* fine-tuned on challenge data — we generated no person-box or wrist labels for this challenge, and they are used zero-shot. Both weight files are shipped unmodified and counted in Table 7. Ultralytics YOLO11 is licensed AGPL-3.0; we therefore distribute the complete corresponding source of the inference package under AGPL-3.0-compatible terms, ship the upstream license and attribution, and make no proprietary-license claim over the detector components.

idea	result	reason on record
depth+IR as <i>primary</i> visual member	−4.7 (subjects) / +0.4 (scenes)	memorizes person/scene appearance
cluster specialists (3 variants)	0.7158 → 0.7128; best stacking +0.16	training signal already saturated
model soup	0.7872 vs 0.7941 alone	members too correlated
transductive self-training	member 0.7544 → 0.7609, ensemble 0.7803 → 0.7658	absorbs the structure prior, loses independence
higher spatial resolution (160/192 px)	per-model +1.3, ensemble gain did not transfer	not a spatial-detail problem
Swin3D-T as an extra member	0.5961 alone, ensemble −0.3	IG65M pretraining is the advantage
Sinkhorn / logit-adjustment	0.7974 → 0.7968	already within 0.061 TV of the training prior
IMU blend	0.3506 alone; 0.7447 → 0.7512	inside noise; 189/3036 clips unreadable
skeleton blend	+0.8	inside noise
radar	no usable signal	—
hard joint scene decoding	0.7076 vs 0.7447	only 79.5% of scenes share exact order
3-bit quantization	chance at every group size	—

Table 13: Ideas rejected during development, with the quantitative or mechanistic reason on record. Numbers are accuracies on the protocol each experiment used.

9 Limitations and Future Work

1. **Structure dependency is the dominant risk.** The decoder’s value is tied to the scripted repeat-take structure. If the organizers re-export data with different filenames, reset timestamps, or run unscripted sessions, the decoder switches off and we are scored on the raw model. We have verified that this degradation is graceful and exact (§4.4), but it is a real drop: from 0.8521 decoded to 0.7770 raw on held-out subjects, a 12–17% relative fall from our 0.90196 private score. *We therefore expect a rerun on structure-free fresh data to land near 0.78, not near 0.90*, and a rerun on Kaggle-like data with the structure present to land within about 5% of the private score. A learned, label-free temporal-consistency prior that does not depend on the thermal frame counter would generalize more cleanly; the guard is a stopgap, not a replacement.
2. **The strideless gain is not cleanly attributed.** As stated in §4.2, the +3.6 points belong to a three-part recipe change. A one-factor ablation is outstanding.
3. **Several intervals rest on few clusters.** The strideless interval resamples 6 subjects. A cluster bootstrap with that few clusters is fragile, and we would not defend the interval’s width, only its sign, which four independent check levels also support.
4. **Hand–object confusions remain.** Residual errors concentrate in close-range hand/object pairs (Table 11). Top-2 accuracy is 9.4 points above top-1 and top-3 is 12.9 above, so the

component	parameters	GMACs / clip
joint body+hands (strideless, 32 frames, 128 px, 2 views)	64.78 M	893.50
depth+IR 160 (strided, 16 frames, 160 px)	63.51 M	156.51
thermal (strided, 16 frames, 128 px)	63.51 M	100.17
IR-only (strided, 16 frames, 128 px)	63.51 M	100.17
YOLO11n person detector (8 probe frames, 640 px)	2.62 M	26.5
YOLO11n-pose wrist detector (8 probe frames, 640 px)	2.87 M	30.2
total, single forward pass	260.80 M	1,307
total with horizontal-flip TTA (as shipped)	260.80 M	2,557 (5.11 TFLOPs)
<i>reference: strided R(2+1)D-34, 32 frames, 128 px</i>	63.51 M	200.33

Table 14: Inference cost per clip. The strideless joint member is 68% of the compute and 46% of the bytes. Flip TTA doubles the video-encoder work; the detector passes are not duplicated. Counting 2 FLOPs per MAC, the shipped pipeline is 5.11 TFLOPs per clip.

quantity	value
total shipped model assets	96,081,702 B (96.08 MB, 96.1% of cap)
GPUs / peak memory at inference	1 / 1.71 GB (RTX 5090)
inference time (405 clips, fresh raw folders)	83.0 s (0.205 s/clip)
training cost, four shipped members	3.8 GPU-hours (1 × RTX 5090)
network at inference	none
training data at inference	none
determinism	exact on fixed GPU/driver; output SHA-256 reproduced

Table 15: Efficiency profile of the primary submission.

correct answer is almost always in a tiny candidate set; a second-stage specialist over that candidate set is the untried idea with the best expected value — though Table 12 is a warning that it may fail for the same reason the first-stage specialists did.

5. **Subject diversity is the real bottleneck.** With only 18 training subjects, the organizer’s eight held-out participants are the honest test of generalization. More diverse subjects — not more epochs or resolution — is where the next points are.

Reproduction

The complete code, weights, configuration, environment (Docker image plus pinned requirements), and the exact Kaggle submission CSVs are provided in the accompanying verification package. The primary submission reproduces byte-for-byte on the public test set from the package with no network access.

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